# PRETZIL JOY VIVARES


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## PROFILE

Detail-oriented Computer Science graduate with experience in software development, system analysis, application testing, and technical documentation. Skilled in developing, testing, and deploying mobile and web-based applications while ensuring functionality, reliability, and user requirements are met. Possesses strong problem-solving, troubleshooting, and communication skills gained through software projects, academic instruction, and technical evaluations. Eager to contribute technical expertise and a commitment to quality in application development, information technology, and software-related roles.

## SKILLS

- **Programming Languages:** C++ | Java | Dart | JavaScript
- **Web & Mobile Development:** Flutter | Node.js | HTML | CSS
- **Databases**: MySQL | MongoDB Atlas | MongoDB Compass
- **Development Tools**: VS Code | Android Studio | Git | GitHub | Heroku CLI
- **Quality Assurance & Software Development:** Software Testing | Quality Assurance | System Analysis | SDLC | Requirements Analysis | Troubleshooting | Debugging
- **System Design & Documentation:** Use Case Diagrams | System Architecture Design | Technical Documentation | Draw.io
- **AI & Productivity Tools:** ChatGPT | GitHub Copilot | Gemini | NotebookLM
- **Collaboration & Productivity:** Google Workspace | Microsoft 365 | Zoom | Google Meet
- **Professional Skills:** Attention to Detail | Analytical Thinking | Problem Solving | Team Collaboration | Communication | Time Management | Leadership

## PROJECTS

- **Property Plant and Equipment Tracking System (PPETS)** — *Flutter, Dart, Node.js, MySQL*

  Developed, tested, and deployed the Property Plant and Equipment Tracking System (PPETS), formerly known as the Asset Distribution Tracking System (ADTS), for the City Accountant's Office of the Local Government Unit (LGU) of General Santos City during a Computer Science Internship. Implemented asset monitoring, transaction management, notification, and record management features using Flutter, Node.js, and MySQL. Improved asset visibility, streamlined administrative workflows, and reduced manual record-keeping through digital process automation.

- **Famie and FamKid (Parental Control Applications)** — *Flutter, Dart, Node.js, MongoDB Atlas, Heroku*

  *Published in The American Journal of Smart Technology and Solutions (AJSTS)*
  *DOI: https://doi.org/10.54536/ajsts.v4i2.4716*
  Proposed and developed the Famie Parental Control System as part of an undergraduate thesis titled "***Optimizing Screen Time Management for Children's Devices: Leveraging Token Bucket and Time-Based Algorithms in a Famie Parental Control System.***" The project consisted of three components: two full-stack mobile applications (Famie for parents and FamKid for children), which I developed, and an administrative web portal developed by a co-researcher. Implemented core features including screen time management using token bucket and time-based algorithms, secure QR code-based device pairing, and centralized monitoring. Contributed to technical documentation, manuscript review, editing, and research validation activities supporting the study.

- **CETE Quiz Game** — *HTML, CSS, Node.js, MongoDB*

  Designed, developed, tested, and deployed a full-stack web application for the College of Engineering and Technology Education (CETE) Intramurals 2024. Developed dynamic question-shuffling algorithms, countdown-based gameplay mechanics, participant record management, and MongoDB database integration to deliver an engaging and interactive quiz experience. The application was later selected as a featured project during the Department Open House, demonstrating full-stack web development, database management, system design, and web application architecture to prospective students and visitors.

- **TicTacToe Mobile Game** — *Flutter, Dart, Node.js, MongoDB*

  Designed, developed, tested, and deployed a mobile application using Flutter and Dart to demonstrate mobile application development, reactive UI rendering, and game state management. Implemented turn-based gameplay logic, dynamic player interactions, win-condition validation, and board reset functionality to deliver an interactive user experience. The application was featured during the HTC Open House 2024 as a live demonstration project, showcasing mobile application development concepts and software engineering principles to prospective students and visitors.

## WORK EXPERIENCE

**Part-Time Faculty**

College of Engineering & Technology Education Department | Junior High School Department

*Holy Trinity College of General Santos City* *Jul 2025-May 2026*

- **College Instructor:** Delivered C++, Java, and Game Development courses through lectures and hands-on programming activities, strengthening students' understanding of software development principles and industry-relevant coding practices.
- **Junior High School Instructor:** Guided students in designing and developing mobile applications using MIT App Inventor, resulting in functional projects while reinforcing software design, testing, and troubleshooting skills.
- **Research Adviser (Grade 12 STEM, SY 2025–2026)**: Mentored students through proposal development, implementation, and project completion, enhancing analytical thinking, documentation, and problem-solving capabilities.
- **Research Title Panelist (Information Technology & Computer Science)**: Evaluated research proposals and provided technical recommendations that improved project feasibility, system planning, and overall research quality.

**Computer Science Intern**

*City Accountant's Office (CAO) — General Santos City* *Feb 2025 – Apr 2025 (256 hrs, ext from 162 hrs)*

- Developed the **Property Plant and Equipment Tracking System (PPETS),** a mobile application designed to support government asset management and monitoring processes as part of the internship program.
- Provided technical support and troubleshooting for computers, printers, software applications, and network connectivity issues, minimizing operational disruptions and ensuring reliable day-to-day office operations.
- Supported administrative and digitalization initiatives through documentation, data management, and technology-related tasks.
- Facilitated employee wellness and engagement activities, including the coordination of daily workplace Zumba sessions.

## EDUCATION

**BACHELOR OF SCIENCE IN COMPUTER SCIENCE** **JUNE 2025**

*Holy Trinity College of General Santos City*

**LEADERSHIP, ACHIEVEMENTS & CAMPUS INVOLVEMENT:**

- **TOP 10 RESEARCH PAPER FINALIST — 3rd General Santos City National Statistics Month Research Forum (2025)**
  Recognized as one of the Top 10 Research Paper Finalists among 95 abstract submissions and selected as one of only four student presenters for the study, "Optimizing Screen Time Management for Children's Devices: Leveraging Token Bucket and Time-Based Algorithms in a Famie Parental Control System," at a forum organized by the Philippine Statistics Authority (PSA) and the Local Government Unit of General Santos City.
- **EXEMPLARY AWARDEE — LGU General Santos City HR Academy (2025)**
  Awarded the Exemplary Awardee distinction for outstanding performance during a 260-hour internship at the City Accountant's Office. Developed the full-stack mobile application PPETS (Property and Procurement Electronic Tracking System) to automate asset management processes, improve record accuracy, and support efficient government property administration.
- **DEPARTMENT PUBLIC INFORMATION OFFICER (PIO) — College of Engineering and Technology Education (SY 2024–2025)**
  Managed departmental communications and information dissemination, serving as a liaison between students, faculty, and student organizations. Supported academic and extracurricular activities through effective communication, coordination, and leadership, earning a Certificate of Appreciation for outstanding service.
- **1ST PLACE — Research Poster Presentation Competition, 13th PSITS Regional Convention 2024**
  Represented Holy Trinity College of General Santos City at the 13th Philippine Society of Information Technology Students (PSITS) Regional Convention 2024 held at SEAIT, Tupi, South Cotabato. Presented the Famie Parental Control System research project, which leverages Time-Based and Token Bucket Algorithms for children's screen time management. Competed against participants from 25 institutions across Region XII and was awarded First Place for excellence in research presentation, technical communication, and innovation.

**TRAININGS & SEMINARS:**

Advanced Database Management Systems | Introduction to Data Analytics | Structuring and Formatting IEEE Papers: Essential Guidelines | Cybersecurity Awareness on National Women's Month 2025 | Safer Internet Day 2025 | ICT Trends in Cybersecurity and Big Data Analytics | AI for Digital Transformation (Including Generative AI) | Unveiling Tomorrow: The Force of Artificial Intelligence and Machine Learning | Artificial Intelligence for Education and Entrepreneurship | STARTUP TALK Series 2025 – Session 3 | Development Conference (DEVCON) 2023.